

Given a matrix of size $N \times M$, print all elements of the matrix in spiral order starting from the top-left corner.

The traversal order is:

Left to Right across the top row
Top to Bottom along the right column
Right to Left across the bottom row
Bottom to Top along the left column

Continue this process until all elements are visited

Input Format

The first line contains two integers N and M, representing the number of rows and columns. The next N lines contain M space-separated integers representing the matrix.

Constraints

$1 \leq N, M \leq 1000$ $-10^9 \leq \text{Matrix}[i][j] \leq 10^9$

Output Format

Print the elements of the matrix in spiral order separated by spaces.

Sample Input 0

Line: 21 Col: 1

3 3
1 2 3
4 5 6
7 8 9

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Run Code

Submit Code

- ☐ Test against custom input

Sample Output 0

1 2 3 6 9 8 7 4 5

Sample Input 1

4 4
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16

Sample Output 1

1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10

```

1  # Enter your code here. Read input from STDIN
2  n, m = map(int, input().split())
3  a = [list(map(int, input().split())) for _ in range(n)]
4  top, bottom = 0, n - 1
5  left, right = 0, m - 1
6  ans = []
7  while top <= bottom and left <= right:
8      for i in range(left, right + 1):
9          ans.append(a[top][i])
10         top += 1
11         for i in range(top, bottom + 1):
12             ans.append(a[i][right])
13         right -= 1
14         if top <= bottom:
15             for i in range(right, left - 1, -1):
16                 ans.append(a[bottom][i])
17             bottom -= 1
18             if left <= right:
19                 for i in range(bottom, top - 1, -1):
20                     ans.append(a[i][left])
21                 left += 1
22 print(*ans)
23

```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✔ Sample Test case 1

Input (stdin)

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1	3 3
2	1 2 3
3	4 5 6
4	7 8 9

Your Output (stdout)

1	1	2	3	6	9	8	7	4	5
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Expected Output

Download

1	1	2	3	6	9	8	7	4	5
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